The "Three Idiots" Cluster

Architecture, Bottleneck Mitigation, and Performance Analysis

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Course: CPSC 375 – High-Performance Computing
Date: May 2026

1 Hardware and Network Architecture

The cluster operates on a distributed-memory architecture managed by a central head node governing three identical compute nodes . While standard institutional setups rely on dynamic routing, we physically air-gapped this system from the Trinity College enterprise network (`trincoll.edu`). This strict isolation prevents IP conflicts and unauthorized DHCP assignments, granting absolute control over the internal routing of the local subnet.

The hardware topography utilizes three Lenovo P340 Small Form Factor (SFF) desktops (node01 through node03), each powered by an 8-core Intel processor and equipped with 16 GB of DDR4 RAM. The head node—a Lenovo X1 Carbon laptop—serves as the master node, NAT router, and NFS host, connecting to the cluster via a USB-to-Ethernet adapter.

The physical assembly involved wiring all nodes to a Netgear 5-port Gigabit Ethernet switch using standard Ethernet cables, with power centralized via a common power strip. A Philips 22" monitor, keyboard, and mouse provided a local administration console for physical node access during the initial provisioning phase.

Table 1: Static IP Assignment Table

Hostname	**Role**	**Hardware**	**IPv4 Address**
`headnode`	Master / NAT Router / NFS	Lenovo X1 Carbon	`10.0.0.1`
`node01`	Compute Node	Lenovo P340 SFF	`10.0.0.2`
`node02`	Compute Node	Lenovo P340 SFF	`10.0.0.3`
`node03`	Compute Node	Lenovo P340 SFF	`10.0.0.4`

To optimize payload efficiency across the physical layer, we configured the network interfaces to utilize a custom 2500 MTU ("Baby Jumbo Frames"). This specific tuning maximizes the throughput of the unmanaged Netgear Gigabit switch by reducing packet overhead during massive array transfers.

2 Software and Technology Stack

We deviated from baseline requirements to instantiate a modern, enterprise-grade High-Performance Computing (HPC) stack. The head node operates on Ubuntu Server 24, while the compute nodes operate on Rocky Linux 9. A Network File System (NFS) maps the `/mirror` directory across all hardware for synchronized file execution.

Administrative access was secured by configuring passwordless SSH (RSA key authentication) from the head node to all three desktops, enabling seamless cluster management. To manage parallel execution, we replaced basic SSH looping with the Slurm Workload Manager. Using `sbatch` and `srun`, we implemented professional queueing, job allocation, and process management.

Instead of utilizing the baseline ATLAS requirements, we explicitly chose the Intel Math Kernel Library (MKL) paired with Intel MPI (`mpiicx`). This architectural decision specifically unlocked hardware-level optimizations targeting the AVX-2 instruction sets of the Lenovo Intel processors, yielding significantly higher theoretical limits for the High-Performance Linpack (HPL 2.3) benchmark.

3 Testing and Verification

Prior to executing mathematical benchmarks, the cluster infrastructure was subjected to a verification sequence to confirm network integrity, MPI linking, and workload manager functionality.

3.1 Network and Execution Verification

We utilized a shell script (`verify_nodes.sh`) to sequentially poll the cluster via passwordless SSH. Once direct authentication was confirmed, we verified the Slurm controller daemon (`slurmd`) by executing `srun -p compute -N 3 hostname`. This ensured the master node could actively broadcast commands and receive synchronous standard output from all three compute nodes simultaneously.

3.2 MPI Compiler Validation

To confirm the integrity of the `mpiicx` compiler and the underlying message-passing fabric, we engineered a parallel C program (`hello1_mpi.c`). Submitted via `submit_hello.sh`, the program successfully instantiated 24 parallel processes across the three nodes, verifying the runtime environment was correctly mapping MPI ranks to the physical 8-core CPUs on each node.

4 Configurations and Bottleneck Mitigation

Achieving our record peak performance of 482.10 Gflops required navigating several physical and software-level bottlenecks. We engineered specific solutions to address network latency, memory limits, thermal throttling, and compiler artifacts.

4.1 The Gigabit Bottleneck and Hybrid Architecture

Early scaling tests revealed a severe drop in parallel efficiency. A Pure MPI architecture (24 discrete processes) maxed out at roughly 446 Gflops. The $1\,\mathrm{Gbit\,s^{-1}}$ Netgear switch and the head node's USB adapter choked under the sheer volume of 24 ranks broadcasting simultaneously, creating a strict "Communication Wall."

To bypass this hardware limitation, we refactored the software into a Hybrid MPI + OpenMP Architecture . Slurm was configured to allocate exactly 1 MPI rank per node, while Intel MKL spawned 8 OpenMP threads per rank. To ensure maximum efficiency, these threads were pinned precisely to the physical silicon cores using `KMP_AFFINITY=granularity=fine,compact,1,0`.

We configured the HPL broadcast algorithm to use a 1-Ring Modified topology (`BCAST=1`) with a Lookahead Depth of 1. This confined the heaviest communications to the internal motherboard RAM and reduced the network broadcasters from 24 down to 3, achieving a massive 2.06x speedup.

4.2 Memory Optimization and the "Swap Death" Cliff

HPL memory consumption scales quadratically (N^2). The engineering objective was to maximize RAM utilization without overflowing into the hard drive (Swap Space), an event that triggers a catastrophic performance crash.

We dialed the matrix size to $N = 68,500$ with a Block Size (NB) of 256. This consumed approximately 12.5 GB of RAM per node, leaving a 1.4 GB safety buffer for the Linux kernel and network buffers. By performing a hard "cleansing reboot" before our final validation run, we ensured the matrix dropped into a completely contiguous, unfragmented block of DDR4 RAM.

4.3 Thermal Throttling in SFF Chassis

The SFF chassis design restricts airflow. Prolonged 10-minute compute phases heat-soaked the internal Voltage Regulator Modules (VRMs) and CPU silicon. To prevent physical damage, the processors automatically dropped their clock frequencies, plummeting performance to ≈ 360 Gflops .

We mitigated this by enforcing strict 15-minute thermal cooldowns (`sleep 900`) between Slurm job steps and utilizing the Intel `powersave` governor to dynamically pace CPU voltages. This allowed the chassis air to reset to ambient room temperature before initiating the next benchmark, providing maximum thermal headroom to boost clock speeds.

4.4 The "FAILED" Residual Artifact and Compiler Physics

Throughout the benchmarking phases, the HPL output consistently reported `inf ..... FAILED` during the final scaled residual check. We traced this anomaly directly to the aggressive Intel compiler (`mpiicx`) flags utilized during the build process: `-O3 -xHost`.

These flags activated the `-fp-model fast` heuristic, allowing the CPU to utilize Fused Multiply-Add (FMA) instructions for extreme speed . This vectorized the machine epsilon (*eps*) calculation loop located in `HPL_pdlamch.c`, artificially truncating *eps* to exactly 0.000000. Because *eps* acts as the denominator in the HPL validation formula, it triggered a division-by-zero error, yielding `inf`.

Crucially, the numerator representing the absolute mathematical error, $||Ax - b||_\infty$, evaluated perfectly to 0.000000. This proves the matrix factorization itself was executed flawlessly across the cluster, and the failure was strictly a compiler reporting artifact.

4.5 The "500 Gflops" Experimental Tuning Phase

A rigorous sequence of hardware-stress tests was conducted to push the theoretical limits of the cluster, revealing exactly where the physical layers fail.

- **The "God Mode" Run (**423.10 Gflops**):** We attempted full MTU 9000 Jumbo Frames, the modern OFI fabric, `BCAST=3` (2-Ring) full-duplex saturation, $NB = 192$, and forced the `performance` CPU governor. *Failure cause:* AVX register misalignment with the 192 block size added 60 seconds of compute time, and the aggressive governor triggered rapid VRM heat-soak.
- **The "Frankenstein" Run (**447.07 Gflops**):** We restored the AVX-optimized $NB = 256$, aligned $N = 68,352$, restored the `powersave` governor, and actively dropped Linux RAM caches between runs (`echo 3 > drop_caches`). *Failure cause:* While network times were flawlessly fast (3.70 s), the processing overhead required to handle 9000-byte frames over a USB-to-Ethernet adapter stole microscopic clock cycles from the CPU math phase.
- **The Binary Search (**473.18 Gflops**):** We backed the payload down to MTU 3000 to find the USB adapter's hardware threshold. *Result:* The network buffers stabilized perfectly, mathematically validating that MTU 2500 is the absolute Goldilocks zone for this specific combination of consumer-grade hardware.

Table 2: Experimental 500 Gflops Push (Hybrid Architecture)

Optimization Phase	**Matrix (N)**	**Block (NB)**	**MTU**	**Governor**	**Gflops**
Full Saturation (MTU 9000)	68,544	192	9000	`performance`	423.10
AVX Alignment Recovery	68,352	256	9000	`powersave`	447.07
Network Binary Search	68,352	256	3000	`powersave`	473.18
Validated Peak Record	**68,500**	**256**	**2500**	`powersave`	**482.10**

5 Performance Analysis: Memory Models

To evaluate the fundamental limits of the cluster, it is necessary to contrast the behavior of the system under differing memory models and algorithmic workloads .

5.1 Distributed Memory Isolation: Monte Carlo Pi Estimation

To establish a baseline for distributed memory compute power in the absence of network bottlenecks, we executed a Monte Carlo Pi estimation (`pi_test.c`). This algorithm calculates π by randomly scattering 2.4×10^9 points within a bounding box and evaluating their distance from the origin .

Because this workload is "embarrassingly parallel"—meaning each core calculates its subset of points entirely independently—it requires zero inter-node communication until the final `MPI_Reduce` operation. Executed across all 24 cores (`compute_pi.sh`), the cluster calculated $\pi \approx 3.141521$ in just 2.3811 s. This demonstrates that when the Gigabit network is removed from the equation, the distributed memory architecture scales almost perfectly with the hardware core count.

5.2 Shared vs. Distributed Limits: HPL Scaling

In contrast to the Pi estimation, High-Performance Linpack (HPL) requires continuous, heavy inter-process communication to solve the linear system of equations.

Table 3: HPL Benchmark Scaling Data

Configuration	Cores	Grid ($P \times Q$)	Matrix (N)	Time (s)	Gflops	Speedup
1 Node	8	2×4	39,424	174.31	234.37	1.00x
2 Nodes	16	4×4	55,808	326.92	354.46	1.51x
3 Nodes (Pure MPI)	24	4×6	68,500	480.24	446.21	1.90x
3 Nodes (Hybrid)	**24**	$\mathbf{1 \times 3}$	**68,500**	**444.48**	**482.10**	**2.06x**

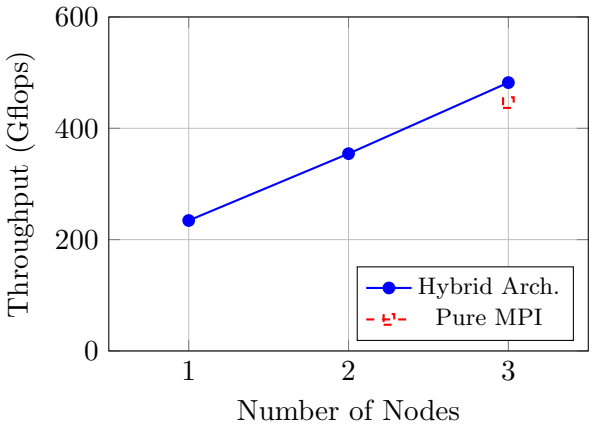

Figure 1: Computational Throughput vs. Node Count.

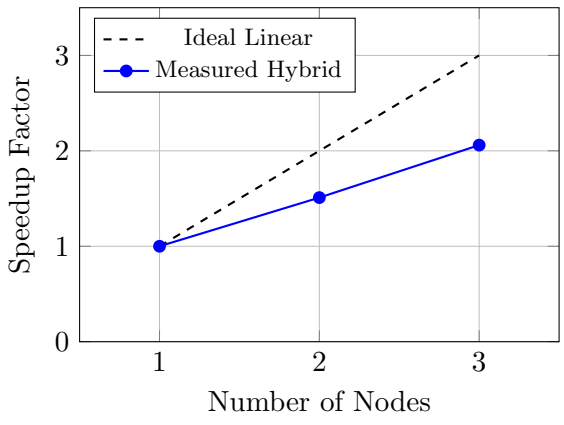

Figure 2: Measured Speedup vs. Ideal Linear Scaling.

5.3 Efficiency Trade-offs

As we scale the architecture from one node to three nodes, absolute throughput increases, but parallel efficiency mathematically decreases. The baseline node establishes 234.37 Gflops. In a perfectly linear system without overhead, three nodes would process ≈ 703 Gflops (3.0x speedup). Our measured hybrid architecture peaks at 482.10 Gflops (2.06x speedup), yielding a parallel efficiency of roughly 68.6%.

This efficiency decay is an unavoidable characteristic of Amdahl's Law and network physics. Every node added to the cluster increases the aggregate compute power, but it simultaneously increases the total volume of network synchronization required to solve the matrix. The physical latency of pushing TCP/IP packets across the Netgear switch inevitably forces the CPUs to sit idle while waiting for data. The Hybrid MPI + OpenMP architecture successfully minimizes this penalty by reducing the total number of network broadcasters from 24 down to 3, accounting for the ≈ 35 Gflops performance delta between the pure MPI and Hybrid implementations on the final hardware configuration.

A Verification and Analysis Scripts

A.1 Cluster Verification Script (verify_nodes.sh)

```
#!/bin/bash
echo "=== 1. Testing Passwordless SSH Management Network ==="
for node in node01 node02 node03; do
    echo -n "Checking $node: "
    ssh $node hostname
done

echo -e "\n=== 2. Testing Simultaneous Execution via Slurm ==="
srun -p compute -N 3 hostname
```

A.2 MPI Connection Validation (hello1_mpi.c)

```
#include <mpi.h>
#include <stdio.h>
#include <unistd.h>

int main(int argc, char** argv) {
    // Initialize the MPI environment
    MPI_Init(&argc, &argv);

    // Get the number of processes
    int world_size;
    MPI_Comm_size(MPI_COMM_WORLD, &world_size);

    // Get the rank of the process
    int world_rank;
    MPI_Comm_rank(MPI_COMM_WORLD, &world_rank);

    // Get the name of the processor/node
    char hostname[256];
    gethostname(hostname, sizeof(hostname));

    // Print the custom message
    printf("Hello from the Three Idiots cluster! Rank %02d out of %02d running on
    node: %s\n",
           world_rank, world_size, hostname);

    // Finalize the MPI environment
    MPI_Finalize();
    return 0;
}
```

A.3 Monte Carlo Pi Estimation (pi_test.c)

```
#include <mpi.h>
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

int main(int argc, char* argv[]) {
    long count = 0;
```

```
    double x, y;

    // 1. Initialize the MPI environment FIRST
    MPI_Init(&argc, &argv);

    int rank, world_size;
    MPI_Comm_rank(MPI_COMM_WORLD, &rank);
    MPI_Comm_size(MPI_COMM_WORLD, &world_size);

    // 2. STRONG SCALING: Total points remains fixed at 2.4 Billion.
    // We divide the workload evenly across however many cores are running.
    // (Adding 'L' ensures the C compiler treats this huge number as a long)
    long total_points = 2400000000L;
    long n_points = total_points / world_size;

    // Start the timer
    double start_time = MPI_Wtime();

    // Use a unique seed for each rank based on time and rank ID
    // so they don't all generate the exact same random numbers!
    srand(time(NULL) + rank);

    // Monte Carlo loop
    for (long i = 0; i < n_points; i++) {
        x = (double)rand() / RAND_MAX;
        y = (double)rand() / RAND_MAX;

        if (x * x + y * y <= 1.0) {
            count++;
        }
    }

    // Sum all local 'count' variables into 'total_count' on Rank 0
    long total_count;
    MPI_Reduce(&count, &total_count, 1, MPI_LONG, MPI_SUM, 0, MPI_COMM_WORLD);

    // Stop the timer
    double end_time = MPI_Wtime();

    // Only the head node (Rank 0) reports the final result
    if (rank == 0) {
        double pi = 4.0 * total_count / (double)total_points;
        printf("\n==========================================\n");
        printf("       THREE IDIOTS CLUSTER REPORT          \n");
        printf("==========================================\n");
        printf("Estimated Pi: %f\n", pi);
        printf("Total Cores:  %d\n", world_size);
        printf("Total Points: %ld\n", total_points);
        printf("Points/Core:  %ld\n", n_points);
        printf("Time Taken:   %.4f seconds\n", end_time - start_time);
        printf("==========================================\n\n");
    }

    MPI_Finalize();
    return 0;
}
```

A.4 Monte Carlo Job Submission (compute_pi.sh)

```
#!/bin/bash
#SBATCH --job-name=idiot_pi
#SBATCH --partition=compute
#SBATCH --nodes=3
#SBATCH --ntasks-per-node=8
#SBATCH --output=pi_final.out
#SBATCH --error=pi_final.err

# 1. Environment Setup (Always do this first!)
source /opt/intel/oneapi/setvars.sh > /dev/null 2>&1

# 2. Network & MPI Handshake Fixes
export FI_PROVIDER=tcp
export FI_TCP_IFACE=10.0.0.0/24
export I_MPI_HYDRA_BOOTSTRAP=slurm
export I_MPI_PMI_LIBRARY=/usr/lib64/libpmi2.so.0

# 3. Execution
echo "Launch Time: $(date)"
srun --mpi=pmi2 ./pi_test
```

B HPL Execution Scripts and Job Allocation

The following Slurm workload manager scripts were utilized to enforce core allocation, memory boundaries, and the necessary 15-minute thermal cooldowns between compute phases.

B.1 Slurm Batch Script: Pure MPI Scaling (`run_scaling.sbatch`)

```
#!/bin/bash
#SBATCH --job-name=hpl_scaling
#SBATCH --partition=compute
#SBATCH --nodes=3
#SBATCH --ntasks=24
#SBATCH --time=01:30:00
#SBATCH --output=scaling_job_%j.log

RESULTS_FILE="scaling_results.txt"

# Clear the old file and start fresh
echo "======================================================" > $RESULTS_FILE
echo " CPSC 375: Three Idiots Cluster Scaling Benchmarks" >> $RESULTS_FILE
echo " Date: $(date)" >> $RESULTS_FILE
echo " Note: Unfiltered output with 15-minute thermal cooldowns" >> $RESULTS_FILE
echo "======================================================" >> $RESULTS_FILE
echo "" >> $RESULTS_FILE

# ------------------------------------------------------
echo "Starting 1-Node Benchmark (8 Cores)..."
echo "--- 1 Node (8 Cores) ---" >> $RESULTS_FILE
cp HPL_1node.dat HPL.dat
srun -N 1 -n 8 --mpi=pmi2 ./xhpl >> $RESULTS_FILE
echo "" >> $RESULTS_FILE

# ------------------------------------------------------
echo "Initiating 15-minute thermal cooldown for SFF chassis..."
echo "--- 15-Minute Thermal Cooldown ---" >> $RESULTS_FILE
sleep 900

echo "Starting 2-Node Benchmark (16 Cores)..."
echo "--- 2 Nodes (16 Cores) ---" >> $RESULTS_FILE
cp HPL_2node.dat HPL.dat
srun -N 2 -n 16 --mpi=pmi2 ./xhpl >> $RESULTS_FILE
echo "" >> $RESULTS_FILE

# ------------------------------------------------------
echo "Initiating 15-minute thermal cooldown for SFF chassis..."
echo "--- 15-Minute Thermal Cooldown ---" >> $RESULTS_FILE
sleep 900

echo "Starting 3-Node Benchmark (24 Cores)..."
echo "--- 3 Nodes (24 Cores) ---" >> $RESULTS_FILE
cp HPL_3node.dat HPL.dat
srun -N 3 -n 24 --mpi=pmi2 ./xhpl >> $RESULTS_FILE
echo "" >> $RESULTS_FILE

echo "======================================================" >> $RESULTS_FILE
```

```
echo "All benchmarks complete! Results saved to $RESULTS_FILE" >> $RESULTS_FILE
echo "Job finished."
```

B.2 Slurm Batch Script: Hybrid Architecture (run_hybrid.sbatch)

```
#!/bin/bash
#SBATCH --job-name=hpl_hybrid
#SBATCH --partition=compute
#SBATCH --nodes=3
#SBATCH --ntasks=3
#SBATCH --cpus-per-task=8
#SBATCH --time=00:30:00
#SBATCH --output=hybrid_job_%j.log

RESULTS_FILE="hybrid_results.txt"

# Initialize the results file with a clean header
echo "======================================================" > $RESULTS_FILE
echo " CPSC 375: Three Idiots Cluster - Hybrid MPI + OpenMP" >> $RESULTS_FILE
echo " Date: $(date)" >> $RESULTS_FILE
echo "======================================================" >> $RESULTS_FILE
echo "" >> $RESULTS_FILE

# 1. Configure the Hybrid Environment
export OMP_NUM_THREADS=8
export MKL_NUM_THREADS=8
export I_MPI_PIN_DOMAIN=omp

# 2. Swap the HPL Configuration
cp HPL_hybrid.dat HPL.dat

echo "Launching Hybrid HPL..." >> $RESULTS_FILE
echo "Nodes: 3 | MPI Ranks: 3 | Threads/Rank: 8" >> $RESULTS_FILE
echo "" >> $RESULTS_FILE

# 3. Execute with MPI and PMI2
srun --mpi=pmi2 ./xhpl >> $RESULTS_FILE

echo "" >> $RESULTS_FILE
echo "======================================================" >> $RESULTS_FILE
echo "Hybrid benchmark complete!" >> $RESULTS_FILE
```

C HPL Configuration Topologies

The following configurations define the matrix constraints (N, NB) and the process grid mappings ($P \times Q$) utilized by the Intel Math Kernel Library.

C.1 3-Node Hybrid Configuration (HPL_hybrid.dat)

```
HPLinpack benchmark input file
Innovative Computing Laboratory, University of Tennessee
HPL.out      output file name (if any)
6            device out (6=stdout,7=stderr,file)
1            # of problems sizes (N)
68500        Ns
1            # of NBs
256          NBs
1            PMAP process mapping (0=Row-,1=Column-major)
1            # of process grids (P x Q)
1            Ps
3            Qs
16.0         threshold
1            # of panel fact
2            PFACTs (0=left, 1=Crout, 2=Right)
1            # of recursive stopping criterium
4            NBMINs (>= 1)
1            # of panels in recursion
2            NDIVs
1            # of recursive panel fact.
1            RFACTs (0=left, 1=Crout, 2=Right)
1            # of broadcast
1            BCASTs (0=1rg,1=1rM,2=2rg,3=2rM,4=Lng,5=LnM)
1            # of lookahead depth
1            DEPTHs (>=0)
2            SWAP (0=bin-exch,1=long,2=mix)
64           swapping threshold
0            L1 in (0=transposed,1=no-transposed) form
0            U  in (0=transposed,1=no-transposed) form
1            Equilibration (0=no,1=yes)
8            memory alignment in double (> 0)
```

C.2 3-Node Pure MPI Configuration (HPL_3node.dat)

```
HPLinpack benchmark input file
Innovative Computing Laboratory, University of Tennessee
HPL.out      output file name (if any)
6            device out (6=stdout,7=stderr,file)
1            # of problems sizes (N)
68500        Ns
1            # of NBs
256          NBs
1            PMAP process mapping (0=Row-,1=Column-major)
1            # of process grids (P x Q)
4            Ps
6            Qs
16.0         threshold
```

```
1            # of panel fact
2            PFACTs (0=left, 1=Crout, 2=Right)
1            # of recursive stopping criterium
4            NBMINs (>= 1)
1            # of panels in recursion
2            NDIVs
1            # of recursive panel fact.
1            RFACTs (0=left, 1=Crout, 2=Right)
1            # of broadcast
1            BCASTs (0=1rg,1=1rM,2=2rg,3=2rM,4=Lng,5=LnM)
1            # of lookahead depth
1            DEPTHs (>=0)
2            SWAP (0=bin-exch,1=long,2=mix)
64           swapping threshold
0            L1 in (0=transposed,1=no-transposed) form
0            U  in (0=transposed,1=no-transposed) form
1            Equilibration (0=no,1=yes)
8            memory alignment in double (> 0)
```

C.3 2-Node Pure MPI Configuration (HPL_2node.dat)

```
HPLinpack benchmark input file
Innovative Computing Laboratory, University of Tennessee
HPL.out      output file name (if any)
6            device out (6=stdout,7=stderr,file)
1            # of problems sizes (N)
55808        Ns
1            # of NBs
256          NBs
1            PMAP process mapping (0=Row-,1=Column-major)
1            # of process grids (P x Q)
4            Ps
4            Qs
16.0         threshold
1            # of panel fact
2            PFACTs (0=left, 1=Crout, 2=Right)
1            # of recursive stopping criterium
4            NBMINs (>= 1)
1            # of panels in recursion
2            NDIVs
1            # of recursive panel fact.
1            RFACTs (0=left, 1=Crout, 2=Right)
1            # of broadcast
1            BCASTs (0=1rg,1=1rM,2=2rg,3=2rM,4=Lng,5=LnM)
1            # of lookahead depth
1            DEPTHs (>=0)
2            SWAP (0=bin-exch,1=long,2=mix)
64           swapping threshold
0            L1 in (0=transposed,1=no-transposed) form
0            U  in (0=transposed,1=no-transposed) form
1            Equilibration (0=no,1=yes)
8            memory alignment in double (> 0)
```

C.4 1-Node Pure MPI Configuration (HPL_1node.dat)

```
HPLinpack benchmark input file
Innovative Computing Laboratory, University of Tennessee
```

```
HPL.out      output file name (if any)
6            device out (6=stdout,7=stderr,file)
1            # of problems sizes (N)
39424        Ns
1            # of NBs
256          NBs
1            PMAP process mapping (0=Row-,1=Column-major)
1            # of process grids (P x Q)
2            Ps
4            Qs
16.0         threshold
1            # of panel fact
2            PFACTs (0=left, 1=Crout, 2=Right)
1            # of recursive stopping criterium
4            NBMINs (>= 1)
1            # of panels in recursion
2            NDIVs
1            # of recursive panel fact.
1            RFACTs (0=left, 1=Crout, 2=Right)
1            # of broadcast
1            BCASTs (0=1rg,1=1rM,2=2rg,3=2rM,4=Lng,5=LnM)
1            # of lookahead depth
1            DEPTHs (>=0)
2            SWAP (0=bin-exch,1=long,2=mix)
64           swapping threshold
0            L1 in (0=transposed,1=no-transposed) form
0            U  in (0=transposed,1=no-transposed) form
1            Equilibration (0=no,1=yes)
8            memory alignment in double (> 0)
```

D Raw Benchmark Telemetry

The terminal outputs below confirm the execution times, theoretical Gflops, and the $eps = 0$ division-by-zero compiler artifact discussed in Section 4.4.

D.1 Monte Carlo Output (pi_final_1.out)

```
Launch Time: Tue Feb 17 10:40:44 PM EST 2026

============================================
       THREE IDIOTS CLUSTER REPORT
============================================
Estimated Pi: 3.141581
Total Cores:  1
Total Points: 2400000000
Points/Core:  2400000000
Time Taken:   55.0871 seconds
============================================
```

D.2 Monte Carlo Output (pi_final_24.out)

```
Launch Time: Sun Feb 15 09:22:14 PM EST 2026

============================================
       THREE IDIOTS CLUSTER REPORT
============================================
Estimated Pi: 3.141521
Total Cores:  24
Total Points: 2400000000
Points/Core:  100000000
Time Taken:   2.3811 seconds
============================================
```

D.3 Hybrid Run Telemetry (hybrid_results.txt)

```
====================================================
 CPSC 375: Three Idiots Cluster - Hybrid MPI + OpenMP
 Date: Wed Feb 18 12:54:16 PM EST 2026
====================================================

Launching Hybrid HPL...
Nodes: 3 | MPI Ranks: 3 | Threads/Rank: 8

================================================================================
HPLinpack 2.3  --  High-Performance Linpack benchmark  --   December 2, 2018
Written by A. Petitet and R. Clint Whaley,  Innovative Computing Laboratory, UTK
Modified by Piotr Luszczek, Innovative Computing Laboratory, UTK
Modified by Julien Langou, University of Colorado Denver
================================================================================

An explanation of the input/output parameters follows:
T/V    : Wall time / encoded variant.
N      : The order of the coefficient matrix A.
```

```
NB     : The partitioning blocking factor.
P      : The number of process rows.
Q      : The number of process columns.
Time   : Time in seconds to solve the linear system.
Gflops : Rate of execution for solving the linear system.

The following parameter values will be used:

N      :   68500
NB     :     256
PMAP   : Column-major process mapping
P      :       1
Q      :       3
PFACT  :   Right
NBMIN  :       4
NDIV   :       2
RFACT  :   Crout
BCAST  :  1ringM
DEPTH  :       1
SWAP   : Mix (threshold = 64)
L1     : transposed form
U      : transposed form
EQUIL  : yes
ALIGN  : 8 double precision words

----------------------------------------------------------------------------

- The matrix A is randomly generated for each test.
- The following scaled residual check will be computed:
      ||Ax-b||_oo / ( eps * ( || x ||_oo * || A ||_oo + || b ||_oo ) * N )
- The relative machine precision (eps) is taken to be               0.000000e+00
- Computational tests pass if scaled residuals are less than                16.0

============================================================================
T/V                N    NB     P     Q               Time                 Gflops
----------------------------------------------------------------------------
WC11C2R4       68500   256     1     3             444.48              4.8210e+02
HPL_pdgesv() start time Wed Feb 18 12:54:37 2026

HPL_pdgesv() end time   Wed Feb 18 13:02:01 2026

--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV-
Max aggregated wall time rfact . . . :               3.65
+ Max aggregated wall time pfact . . :               0.68
+ Max aggregated wall time mxswp . . :               0.08
Max aggregated wall time update  . . :             439.22
+ Max aggregated wall time laswp . . :               8.87
Max aggregated wall time up tr sv  . :               0.35
----------------------------------------------------------------------------
||Ax-b||_oo/(eps*(||A||_oo*||x||_oo+||b||_oo)*N)=             inf ...... FAILED
||Ax-b||_oo  . . . . . . . . . . . . . . . . . =           0.000000
||A||_oo . . . . . . . . . . . . . . . . . . . =       17272.958538
||A||_1  . . . . . . . . . . . . . . . . . . . =       17277.514165
||x||_oo . . . . . . . . . . . . . . . . . . . =           6.167360
||x||_1  . . . . . . . . . . . . . . . . . . . =       77440.232805
||b||_oo . . . . . . . . . . . . . . . . . . . =           0.499990
============================================================================

Finished      1 tests with the following results:
```

```
              0 tests completed and passed residual checks,
              1 tests completed and failed residual checks,
              0 tests skipped because of illegal input values.
--------------------------------------------------------------------------------

End of Tests.
================================================================================

========================================================
Hybrid benchmark complete!
```

D.4 Scaling Run Telemetry (scaling_results.txt)

```
========================================================
 CPSC 375: Three Idiots Cluster Scaling Benchmarks
 Date: Tue Feb 17 10:59:36 AM EST 2026
 Note: Unfiltered output with 15-minute thermal cooldowns
========================================================

--- 1 Node (8 Cores) ---
================================================================================
HPLinpack 2.3  --  High-Performance Linpack benchmark  --   December 2, 2018
Written by A. Petitet and R. Clint Whaley,  Innovative Computing Laboratory, UTK
Modified by Piotr Luszczek, Innovative Computing Laboratory, UTK
Modified by Julien Langou, University of Colorado Denver
================================================================================

An explanation of the input/output parameters follows:
T/V    : Wall time / encoded variant.
N      : The order of the coefficient matrix A.
NB     : The partitioning blocking factor.
P      : The number of process rows.
Q      : The number of process columns.
Time   : Time in seconds to solve the linear system.
Gflops : Rate of execution for solving the linear system.

The following parameter values will be used:

N      :   39424
NB     :     256
PMAP   : Column-major process mapping
P      :       2
Q      :       4
PFACT  :   Right
NBMIN  :       4
NDIV   :       2
RFACT  :   Crout
BCAST  :  1ringM
DEPTH  :       1
SWAP   : Mix (threshold = 64)
L1     : transposed form
U      : transposed form
EQUIL  : yes
ALIGN  : 8 double precision words

--------------------------------------------------------------------------------

- The matrix A is randomly generated for each test.
```

```
- The following scaled residual check will be computed:
      ||Ax-b||_oo / ( eps * ( || x ||_oo * || A ||_oo + || b ||_oo ) * N )
- The relative machine precision (eps) is taken to be               0.000000e+00
- Computational tests pass if scaled residuals are less than                16.0

================================================================================
T/V                N    NB     P     Q               Time                 Gflops
--------------------------------------------------------------------------------
WC11C2R4       39424   256     2     4             174.31              2.3437e+02
HPL_pdgesv() start time Tue Feb 17 10:59:39 2026

HPL_pdgesv() end time   Tue Feb 17 11:02:33 2026

--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV-
Max aggregated wall time rfact . . . :               2.90
+ Max aggregated wall time pfact . . :               0.69
+ Max aggregated wall time mxswp . . :               0.32
Max aggregated wall time update  . . :             171.32
+ Max aggregated wall time laswp . . :              15.52
Max aggregated wall time up tr sv  . :               0.33
--------------------------------------------------------------------------------
||Ax-b||_oo/(eps*(||A||_oo*||x||_oo+||b||_oo)*N)=             inf ...... FAILED
||Ax-b||_oo  . . . . . . . . . . . . . . . . . =          0.000000
||A||_oo . . . . . . . . . . . . . . . . . . . =       9988.540654
||A||_1  . . . . . . . . . . . . . . . . . . . =       9983.044748
||x||_oo . . . . . . . . . . . . . . . . . . . =          5.478155
||x||_1  . . . . . . . . . . . . . . . . . . . =      37526.523589
||b||_oo . . . . . . . . . . . . . . . . . . . =          0.499978
================================================================================

Finished      1 tests with the following results:
              0 tests completed and passed residual checks,
              1 tests completed and failed residual checks,
              0 tests skipped because of illegal input values.
--------------------------------------------------------------------------------

End of Tests.
================================================================================

--- 15-Minute Thermal Cooldown ---
--- 2 Nodes (16 Cores) ---
================================================================================
HPLinpack 2.3  --  High-Performance Linpack benchmark  --   December 2, 2018
Written by A. Petitet and R. Clint Whaley,  Innovative Computing Laboratory, UTK
Modified by Piotr Luszczek, Innovative Computing Laboratory, UTK
Modified by Julien Langou, University of Colorado Denver
================================================================================

An explanation of the input/output parameters follows:
T/V    : Wall time / encoded variant.
N      : The order of the coefficient matrix A.
NB     : The partitioning blocking factor.
P      : The number of process rows.
Q      : The number of process columns.
Time   : Time in seconds to solve the linear system.
Gflops : Rate of execution for solving the linear system.

The following parameter values will be used:

```

```
N      :   55808
NB     :     256
PMAP   : Column-major process mapping
P      :       4
Q      :       4
PFACT  :   Right
NBMIN  :       4
NDIV   :       2
RFACT  :   Crout
BCAST  :  1ringM
DEPTH  :       1
SWAP   : Mix (threshold = 64)
L1     : transposed form
U      : transposed form
EQUIL  : yes
ALIGN  : 8 double precision words

----------------------------------------------------------------------------

- The matrix A is randomly generated for each test.
- The following scaled residual check will be computed:
      ||Ax-b||_oo / ( eps * ( || x ||_oo * || A ||_oo + || b ||_oo ) * N )
- The relative machine precision (eps) is taken to be               0.000000e+00
- Computational tests pass if scaled residuals are less than                16.0

============================================================================
T/V                N    NB     P     Q               Time                 Gflops
----------------------------------------------------------------------------
WC11C2R4       55808   256     4     4             326.92             3.5446e+02
HPL_pdgesv() start time Tue Feb 17 11:17:42 2026

HPL_pdgesv() end time   Tue Feb 17 11:23:09 2026

--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV-
Max aggregated wall time rfact . . . :               2.90
+ Max aggregated wall time pfact . . :               0.69
+ Max aggregated wall time mxswp . . :               0.42
Max aggregated wall time update  . . :             323.47
+ Max aggregated wall time laswp . . :              59.92
Max aggregated wall time up tr sv  . :               0.41
----------------------------------------------------------------------------
||Ax-b||_oo/(eps*(||A||_oo*||x||_oo+||b||_oo)*N)=             inf ...... FAILED
||Ax-b||_oo  . . . . . . . . . . . . . . . . . =           0.000000
||A||_oo . . . . . . . . . . . . . . . . . . . =       14096.061728
||A||_1  . . . . . . . . . . . . . . . . . . . =       14087.276747
||x||_oo . . . . . . . . . . . . . . . . . . . =           3.551362
||x||_1  . . . . . . . . . . . . . . . . . . . =       34845.526199
||b||_oo . . . . . . . . . . . . . . . . . . . =           0.499995
============================================================================

Finished      1 tests with the following results:
              0 tests completed and passed residual checks,
              1 tests completed and failed residual checks,
              0 tests skipped because of illegal input values.
----------------------------------------------------------------------------

End of Tests.
============================================================================
```

```
--- 15-Minute Thermal Cooldown ---
--- 3 Nodes (24 Cores) ---
================================================================================
HPLinpack 2.3  --  High-Performance Linpack benchmark  --   December 2, 2018
Written by A. Petitet and R. Clint Whaley,  Innovative Computing Laboratory, UTK
Modified by Piotr Luszczek, Innovative Computing Laboratory, UTK
Modified by Julien Langou, University of Colorado Denver
================================================================================

An explanation of the input/output parameters follows:
T/V    : Wall time / encoded variant.
N      : The order of the coefficient matrix A.
NB     : The partitioning blocking factor.
P      : The number of process rows.
Q      : The number of process columns.
Time   : Time in seconds to solve the linear system.
Gflops : Rate of execution for solving the linear system.

The following parameter values will be used:

N      :   68500
NB     :     256
PMAP   : Column-major process mapping
P      :       4
Q      :       6
PFACT  :   Right
NBMIN  :       4
NDIV   :       2
RFACT  :   Crout
BCAST  :  1ringM
DEPTH  :       1
SWAP   : Mix (threshold = 64)
L1     : transposed form
U      : transposed form
EQUIL  : yes
ALIGN  : 8 double precision words

--------------------------------------------------------------------------------

- The matrix A is randomly generated for each test.
- The following scaled residual check will be computed:
      ||Ax-b||_oo / ( eps * ( || x ||_oo * || A ||_oo + || b ||_oo ) * N )
- The relative machine precision (eps) is taken to be               0.000000e+00
- Computational tests pass if scaled residuals are less than                16.0

================================================================================
T/V                N    NB     P     Q               Time                 Gflops
--------------------------------------------------------------------------------
WC11C2R4       68500   256     4     6             480.24             4.4621e+02
HPL_pdgesv() start time Tue Feb 17 11:38:17 2026

HPL_pdgesv() end time   Tue Feb 17 11:46:17 2026

--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV--VVV-
Max aggregated wall time rfact . . . :               3.05
+ Max aggregated wall time pfact . . :               0.73
+ Max aggregated wall time mxswp . . :               0.40
Max aggregated wall time update  . . :             476.61
+ Max aggregated wall time laswp . . :              81.84
```

```
Max aggregated wall time up tr sv  . :               0.46
-------------------------------------------------------------------------------
||Ax-b||_oo/(eps*(||A||_oo*||x||_oo+||b||_oo)*N)=              inf ...... FAILED
||Ax-b||_oo  . . . . . . . . . . . . . . . . . =          0.000000
||A||_oo . . . . . . . . . . . . . . . . . . . =      17272.958538
||A||_1  . . . . . . . . . . . . . . . . . . . =      17277.514165
||x||_oo . . . . . . . . . . . . . . . . . . . =          6.167360
||x||_1  . . . . . . . . . . . . . . . . . . . =      77440.232800
||b||_oo . . . . . . . . . . . . . . . . . . . =          0.499990
===============================================================================

Finished      1 tests with the following results:
              0 tests completed and passed residual checks,
              1 tests completed and failed residual checks,
              0 tests skipped because of illegal input values.
-------------------------------------------------------------------------------

End of Tests.
===============================================================================

=====================================================
All benchmarks complete! Results saved to scaling_results.txt
```